

Triangles, radians and sectors - Set 2 - Solutions

Specification MM4.1, MM4.2, M5.16.

Answer key

Question	Answer	Topic
1	A	Triangles, radians and sectors
2	E	Triangles, radians and sectors
3	D	Triangles, radians and sectors
4	D	Triangles, radians and sectors
5	C	Triangles, radians and sectors

Worked solutions

Question 1 TRI-11 · Triangles, radians and sectors · spec MM4.2

A circle has radius r . A sector of angle θ radians has the same area as the segment cut off by a chord that subtends an angle of $\frac{\pi}{2}$ at the centre. Find θ .

Answer: A

Segment area $= \frac{1}{2}r^2(\frac{\pi}{2} - \sin \frac{\pi}{2}) = \frac{1}{2}r^2(\frac{\pi}{2} - 1)$. Sector area $\frac{1}{2}r^2\theta$. So $\theta = \frac{\pi}{2} - 1$.

Full method - how and why

Equate two area expressions. The radius cancels, leaving an equation for θ alone.

1 Segment area for a right angle at the centre: $\frac{1}{2}r^2(\frac{\pi}{2} - \sin \frac{\pi}{2}) = \frac{1}{2}r^2(\frac{\pi}{2} - 1)$.

why: Segment = sector – triangle $= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin \theta$, and $\sin \frac{\pi}{2} = 1$.

2 Sector area for angle θ : $\frac{1}{2}r^2\theta$.

why: Same radius, different angle.

3 Set them equal: $\frac{1}{2}r^2\theta = \frac{1}{2}r^2(\frac{\pi}{2} - 1)$.

why: Both areas are in the same circle, so r is common.

4 Cancel $\frac{1}{2}r^2$: $\theta = \frac{\pi}{2} - 1$.

why: The answer is independent of r , which is why the question never tells you its value.

Where it goes wrong

- Using $\sin \frac{\pi}{2} = 0$ and answering $\frac{\pi}{2}$.
- Forgetting to cancel and leaving an r in the answer.

Takeaway When two areas in the same circle are equated, r cancels - the answer depends only on the angles.

Question 2 TRI-12 · Triangles, radians and sectors · spec MM4.1

A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?

- I The angle opposite c is obtuse.
- II c is the longest side.
- III The triangle cannot be isosceles.

Answer: E

Cosine rule: $\cos C = \frac{a^2 + b^2 - c^2}{2ab} < 0$, so C is obtuse (I). The largest angle is opposite the longest side, so c is longest (II). III is false: $a = b = 1, c = 1.5$ works.

Full method - how and why

A condition on the sides. The cosine rule converts it into a statement about an angle; then test the third claim with a counterexample.

1 I: $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$, and $a^2 + b^2 < c^2$ makes the numerator negative, so $\cos C < 0$ and C is obtuse. **True.**
why: The denominator $2ab$ is positive, so the sign of $\cos C$ is the sign of $a^2 + b^2 - c^2$.

2 II: a triangle can have at most one obtuse angle, and the largest angle is opposite the largest side. Since C is obtuse it is the largest angle, so c is the longest side. **True.**
why: Two obtuse angles would already exceed 180° .

3 III: try $a = b = 1$ and $c = 1.5$: then $a^2 + b^2 = 2 < 2.25 = c^2$ ✓ and the triangle is isosceles. **False.**
why: A single counterexample kills the claim. Check the triangle inequality too: $1 + 1 > 1.5$ ✓.

4 I and II only.
why: Match to the option list.

Where it goes wrong

- Assuming an obtuse triangle cannot be isosceles.
- Forgetting to check the triangle inequality when constructing a counterexample.

Takeaway $a^2 + b^2 < c^2 \Leftrightarrow$ the angle opposite c is obtuse; obtuse triangles can still be isosceles.

Question 3 TRI-14 · Triangles, radians and sectors · spec MM4.1

A vertical pole BT stands on horizontal ground. From a point A on the ground the angle of elevation of T is 30° ; from a point C on the ground, 20 m closer to the pole than A and in line with A and B , the angle of elevation is 60° . Find the height of the pole.

Answer: D

Let the height be h . $AB = h\sqrt{3}$ and $CB = \frac{h}{\sqrt{3}}$. $AB - CB = 20 \Rightarrow h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} = 20 \Rightarrow h = 10\sqrt{3}$.

Full method - how and why

Two right-angled triangles sharing the same height. Express both base lengths in terms of h and use the given difference.

- From A : $\tan 30^\circ = \frac{h}{AB}$, so $AB = \frac{h}{\tan 30^\circ} = h\sqrt{3}$.
why: $\tan 30^\circ = \frac{1}{\sqrt{3}}$, so dividing by it multiplies by $\sqrt{3}$.
- From C : $\tan 60^\circ = \frac{h}{CB}$, so $CB = \frac{h}{\tan 60^\circ} = \frac{h}{\sqrt{3}}$.
why: $\tan 60^\circ = \sqrt{3}$. The steeper angle means the shorter base - a good sanity check.
- Use the 20 m gap: $AB - CB = h\sqrt{3} - \frac{h}{\sqrt{3}} = h\left(\frac{3-1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} = 20$.
why: Common denominator $\sqrt{3}$: $\sqrt{3} = \frac{3}{\sqrt{3}}$.
- Solve: $h = \frac{20\sqrt{3}}{2} = 10\sqrt{3}$ m.
why: Multiply both sides by $\sqrt{3}$ and halve.

Where it goes wrong

- Adding the bases instead of subtracting (the closer point has the **shorter** base).
- Using $\tan 30^\circ = \sqrt{3}$, which swaps the two triangles.

Takeaway Two elevation angles to the same object: write each horizontal distance in terms of the height and use their difference.

Question 4 TRI-15 · Triangles, radians and sectors · spec MM4.1

Triangle ABC has $AB = 7$, $BC = 8$, $CA = 9$. Find the area of the triangle.

Answer: D

$\cos A = \frac{49 + 81 - 64}{2 \cdot 7 \cdot 9} = \frac{66}{126} = \frac{11}{21}$, so $\sin A = \frac{\sqrt{441 - 121}}{21} = \frac{\sqrt{320}}{21} = \frac{8\sqrt{5}}{21}$. Area = $\frac{1}{2} \cdot 7 \cdot 9 \cdot \frac{8\sqrt{5}}{21} = 12\sqrt{5}$.

Full method - how and why

Three sides, area wanted. Find a cosine, convert to a sine, then use $\frac{1}{2}ab\sin C$ - all without a calculator.

- Find $\cos A$ where A is between the sides 7 and 9: $\cos A = \frac{49 + 81 - 64}{2(7)(9)} = \frac{66}{126} = \frac{11}{21}$.
why: The side opposite A is $BC = 8$, so that is the one subtracted.
- Convert to $\sin A$ using $\sin^2 + \cos^2 = 1$: $\sin A = \sqrt{1 - \frac{121}{441}} = \frac{\sqrt{320}}{21}$.
why: Keep the denominator 21 throughout: $1 = \frac{441}{441}$.
- Simplify the surd: $\sqrt{320} = \sqrt{64 \times 5} = 8\sqrt{5}$, so $\sin A = \frac{8\sqrt{5}}{21}$.
why: Pull out the largest square factor.
- Area = $\frac{1}{2}(7)(9)\sin A = \frac{63}{2} \times \frac{8\sqrt{5}}{21} = 12\sqrt{5}$.
why: $\frac{63 \times 8}{2 \times 21} = \frac{504}{42} = 12$.

Where it goes wrong

- Taking $\sin A$ positive is right (angles in a triangle are between 0° and 180°), but forgetting to square $\cos A$ before subtracting.
- Arithmetic slips in $\sqrt{320}$, e.g. $16\sqrt{5}$, doubling the area.

Takeaway Three sides $\rightarrow \cos$ from the cosine rule $\rightarrow \sin$ from the Pythagorean identity $\rightarrow$ area from $\frac{1}{2}ab\sin C$.

Question 5 TRI-16 · Triangles, radians and sectors · spec MM4.2

Two circles of radius 2 have their centres a distance 2 apart. Find the area of the region common to both circles.

Answer: C

The common chord subtends $\frac{2\pi}{3}$ at each centre (equilateral triangles of side 2). The overlap is two segments:
 $2 \times \frac{1}{2} \cdot 4 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right) = 4 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) = \frac{8\pi}{3} - 2\sqrt{3}$.

Full method - how and why

The overlap of two circles is two segments. The geometry of the centres gives the angle, then the segment formula applies twice.

- 1 Let the circles meet at P and Q . Each centre, together with P and Q , forms a triangle with sides 2, 2 and the chord PQ .
why: Both radii are 2 and the centres are 2 apart, so joining a centre to P and to the other centre gives an equilateral triangle of side 2.
- 2 That equilateral triangle shows the half-angle at a centre is 60° , so the chord subtends $\frac{2\pi}{3}$ (i.e. 120°) at each centre.
why: Two equilateral triangles back to back, one for P and one for Q .
- 3 Compute one segment: $\frac{1}{2}r^2(\theta - \sin \theta) = \frac{1}{2}(4) \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) = \frac{4\pi}{3} - \sqrt{3}$.
why: $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$.
- 4 The overlap consists of two such segments (one from each circle): $2 \left(\frac{4\pi}{3} - \sqrt{3} \right) = \frac{8\pi}{3} - 2\sqrt{3}$.
why: By symmetry the two segments are congruent.

Where it goes wrong

- Counting only one segment, giving $\frac{4\pi}{3} - \sqrt{3}$ - an offered option.
- Using $\theta = \frac{\pi}{3}$ (the half-angle) instead of $\frac{2\pi}{3}$.

Takeaway Overlapping circles: the lens is two segments; get the central angle from the triangle formed by the two centres and an intersection point.